Assignment 5 — Optimization and LR Schedulers

Raisin · 7–20–3–1 network · L1 λ = 0.7

Method

Stratified 60/20/20 split; train-only standardization. Same initial weights and batch order; 5,000 epochs; batch size 64. Initial LR: GD/Momentum 0.1, Adam 0.01. Reported loss is binary cross-entropy without the L1 penalty.

Final validation results

Optimizer	Scheduler	Loss	Accuracy
ADAM	constant	0.5515	85.56%
GD	tutorial decay	0.5534	85.56%
ADAM	step	0.5582	86.11%
MOMENTUM	tutorial decay	0.5588	86.67%
MOMENTUM	step	0.5640	86.67%
MOMENTUM	constant	0.5706	86.11%
GD	step	0.5731	86.67%
ADAM	tutorial decay	0.5774	84.44%
GD	constant	0.5821	85.56%

Findings

Tutorial decay reduced validation loss for GD (0.5821 → 0.5534) and Momentum (0.5706 → 0.5588), but increased it for Adam (0.5515 → 0.5774). Step decay gave GD and Momentum 86.67% validation accuracy. Adam with a constant rate had the lowest validation loss. Selected using validation loss only, it reached 87.22% accuracy on the held-out test set.

Loss and accuracy rank models differently because cross-entropy considers prediction confidence. The train/validation gap suggests overfitting; one split does not establish a universal best method.

The tutorial's 'exponential' formula is inverse-time: $LR_0/(1+0.001\times\text{epoch})$.

Step decay: $LR_0\times 0.7^{\text{floor}(\text{epoch}/1000)}$.

Data: Çinar, Koklu & Tasdemir (2020), UCI Raisin, DOI: 10.24432/C5660T (CC BY 4.0).

Training and validation loss

